

Carter Mitchell Betts

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EDUCATION

Ohio Northern University – *Ada, Ohio* **August 2023 – Expected May 2027**
• **Bachelor of Science in Mechanical Engineering** – Robotics Concentration, Computer Science Minor | 3.96 GPA

PROFESSIONAL EXPERIENCE

Evans Research Fellow – *Ohio Northern University, Ada, Ohio* **May 2026 – August 2026**

- Optimized COLMAP photogrammetry pipeline for patient-specific surgical planning, achieving model surface noise repeatable within 0.257 mm across 27 trials
- Established metric scale for photogrammetric reconstruction by automatically measuring known fixed geometry and computing required scale factor, with a coefficient of variation of 0.6% across 14 reconstructions
- Built and trained a PointNet++ deep learning segmentation model to automatically identify hand webspace in patient scans with a mean intersection-over-union (mIoU) of 0.774 under 5-fold cross-validation
- Developed Python model of collagen fiber networks to relate fiber crimp state to skin elastic modulus, validated against published experimental data

Process Innovation Engineering Intern – *Howmet Aerospace, Cleveland, Ohio* **May 2025 – August 2025**

- Cut projected lubrication costs by \$400,000+ on aluminum forging press through hands-on testing of lubrication dilution and delivery method
- Engineered a custom viscometer from concept to completion, combining specialized hardware design, circuit development, and embedded programming to reduce calculation turnaround from roughly 7 days to less than 1 hour

Product Integrity Engineering Intern – *Howmet Aerospace, Cleveland, Ohio* **May 2024 – August 2024**

- Developed R scripts to automate critical lifecycle calculations and figure generation, reducing analysis and reporting turnaround by ~70%
- Automated strain gauge drilling tests by designing, programming, and assembling a custom robotic testing rig, eliminating manual measurement error

RESEARCH EXPERIENCE

Ohio Space Grant Consortium STEM Scholar **August 2026 – May 2027**

- Designing low-footprint, smart microgravity exercise equipment for deep space missions on small crew vehicles
- Selected to present research findings at 35th Annual OSGC Student Research Symposium

Undergraduate Research Assistant – Multilayered Artificial Nail Biomechanics **March 2026 – May 2026**

- Developed ANSYS finite element model to simulate loading on complex fingernail assembly
- Performed convergence study to optimize element sizing and verify solution independence from mesh density

Undergraduate Research Assistant – Syndactyly Surgical Planning **January 2024 – Present**

- Performed finite element analysis on hand webspace through the use of ANSYS Parametric Design Language (APDL)
- Built a Python-based embedded control and imaging system to drive a stepper motor, synchronize multi-camera capture, and reconstruct 3D geometry for surgical optimization

PUBLICATIONS

- “Biomechanical Analysis of Multilayered Artificial Nail Systems: A Survey-Informed Finite Element Approach” – second author, ASME IMECE 2026, Paper No. IMECE2026-192805 (Accepted)
- “A Photogrammetry-Based Approach for Patient-Specific Modeling in Syndactyly Surgery” – fourth author, ASME IMECE 2025, Paper No. IMECE2025-166019; presented full technical session, Memphis, TN

LEADERSHIP

American Society of Mechanical Engineers – Vice President **April 2024 – April 2026**

- Led weekly meetings of 40+ engineering students and organized a cross-organizational fundraiser raising over \$1,700

Sigma Pi International – Brotherhood Committee Head **August 2025 – August 2026**

- Coordinated chapter-wide events and managed the semester event calendar for 30+ members

TECHNICAL SKILLS

Languages: Python, Java, MATLAB, C++, ANSYS Parametric Design Language (APDL), R

Development: Git/GitHub, Eclipse, VS Code, Tkinter, GUI application development, object-oriented design

Simulation & CAD: ANSYS Mechanical, pyMAPDL, SolidWorks, OnShape

Machine Learning & 3D Vision: OpenCV, PyTorch, PointNet++, COLMAP, PyVista, PyMeshLab, 3DF Zephyr

Embedded Systems: Arduino, Raspberry Pi, Programmable Logic Controllers (PLC), LabVIEW, circuit integration